SE8335

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MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)

UG Programme (Honors) Degree October 2025 Regular Examinations (UG III Semester)

Third Semester

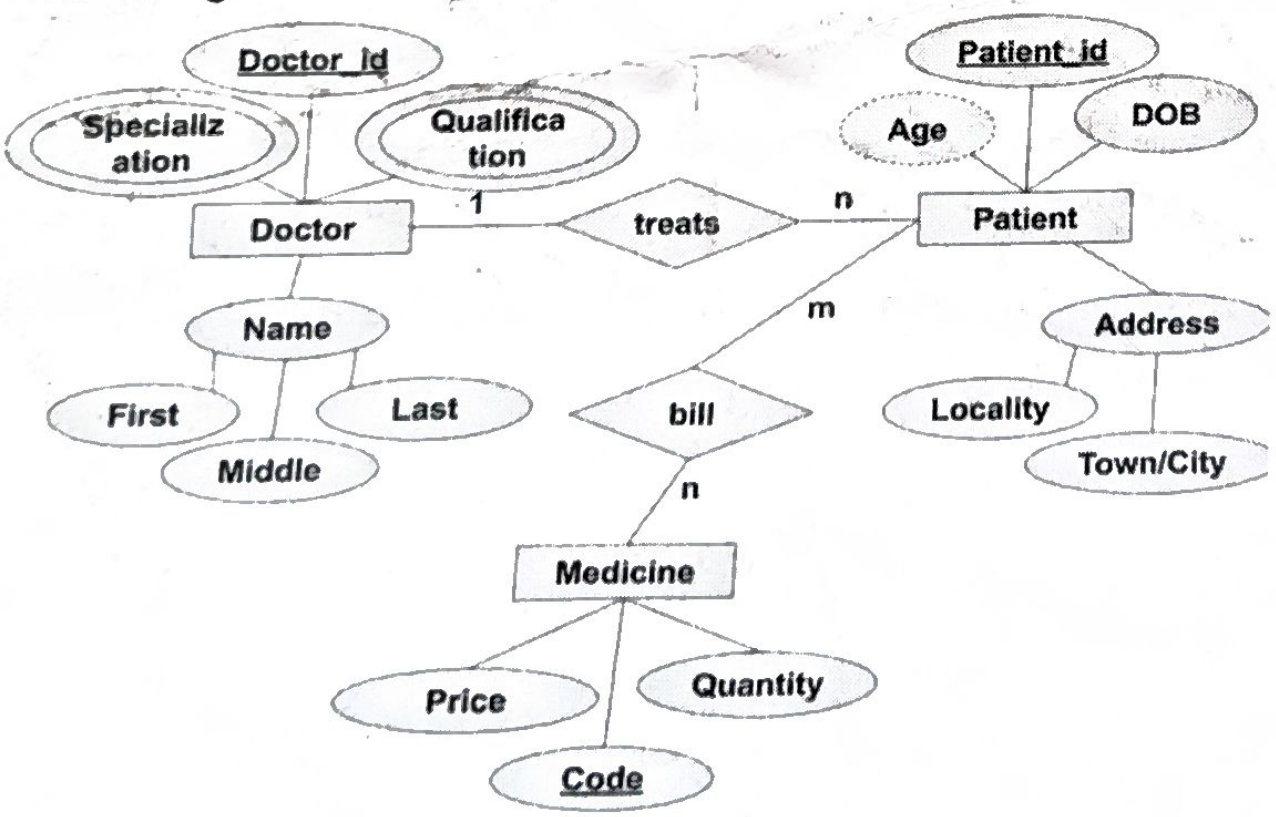
243CCRUBC202 Database Management Systems

Time: Maximum Marks: 60

Answer one question from each bunch

BUNCH I

I.A With the given ER diagram,



Hospital Management System

I.A.1 Identify and classify the different types of attributes in the Hospital Management System ER diagram, and provide suitable explanations for each type. (CO1:5 Marks)

I.A.2 Examine the relationships in the given Hospital Management System ER diagram. For each relationship, identify its type (one-to-one, one-to-many, or many-to-many) and explain its cardinality with justification. (CO1:5 Marks)

I.A.3 Based on the entities shown in the Hospital Management System ER diagram, distinguish between strong and weak entity types. Are there any weak entities in this model? Justify your answer with appropriate reasoning. (CO1:5 Marks)

OR

I.B.1 Explain the different types of database languages in DBMS and analyze their roles in database operations with suitable examples. (CO1:8 Marks)

I.B.2 Describe the relational data model and demonstrate its use with examples to show how data is organized, stored, and retrieved. (CO1:7 Marks)

BUNCH II

II.A.1 Identify the relational algebra operations from set theory and discuss them with examples. (CO2:9 Marks)

II.A.2 Illustrate a relation from any database by drawing it and showing an instance. Clearly indicate the attributes and tuples in the relation. (CO2:6 Marks)

OR

II.B.1 Describe the significance of entity integrity and referential integrity constraints in a relational model (CO2:7.5 Marks)

II.B.2 Explain the following terms in view of a relational database.

- Relation
- Tuple
- Attribute

(CO2:7.5 Marks)

BUNCH III

III.A Differentiate between the following in SQL, providing suitable examples for each:

i) LEFT JOIN and RIGHT JOIN
ii) CHAR and VARCHAR
iii) TRUNCATE TABLE and DROP TABLE

(CO3:15 Marks)

OR

III.B Consider the relations.
Customer(CID, CName, City)
Supplier(SID, SName, City)

Write SQL queries using set operations for the following:
a) Create the table and insert 3 records into the table.

b) Retrieve the names of all people who are either customers or suppliers.

c) Find the names of people who are both customers and suppliers.

d) List the names of customers who are not suppliers.

(CO3:15 Marks)

BUNCH IV

IV.A.1 Explain the concept of functional dependency and illustrate its role in achieving database normalization. Show with examples how understanding functional dependencies helps in designing a consistent and efficient database.

(CO4:9 Marks)

IV.A.2 Consider the following relation schema for a Student database: Student(StudentID, Name, Course, Instructor, Grade). Identify and write all possible functional dependencies (FDs) in this relation.

(CO4:6 Marks)

OR

IV.B NoSQL databases are used in Big Data applications for scalability and flexibility.

IV.B.1 Explain the features and applications of NoSQL databases, and provide a suitable example to illustrate their use.

(CO4:7.5 Marks)

IV.B.2 Compare BASE and ACID properties with one real-world example each.

(CO4:7.5 Marks)

CO1	Explain DBMS concepts, data models and ER model.
CO2	Demonstrate relational data model and apply relational algebra operations.
CO3	Apply SQL for database management.
CO4	Demonstrate Normalization Techniques, Transaction Management Principles and NoSQL database concepts
CO5	Implement SQL query, and administer MongoDB databases.

---------- DIEDIIT ----------